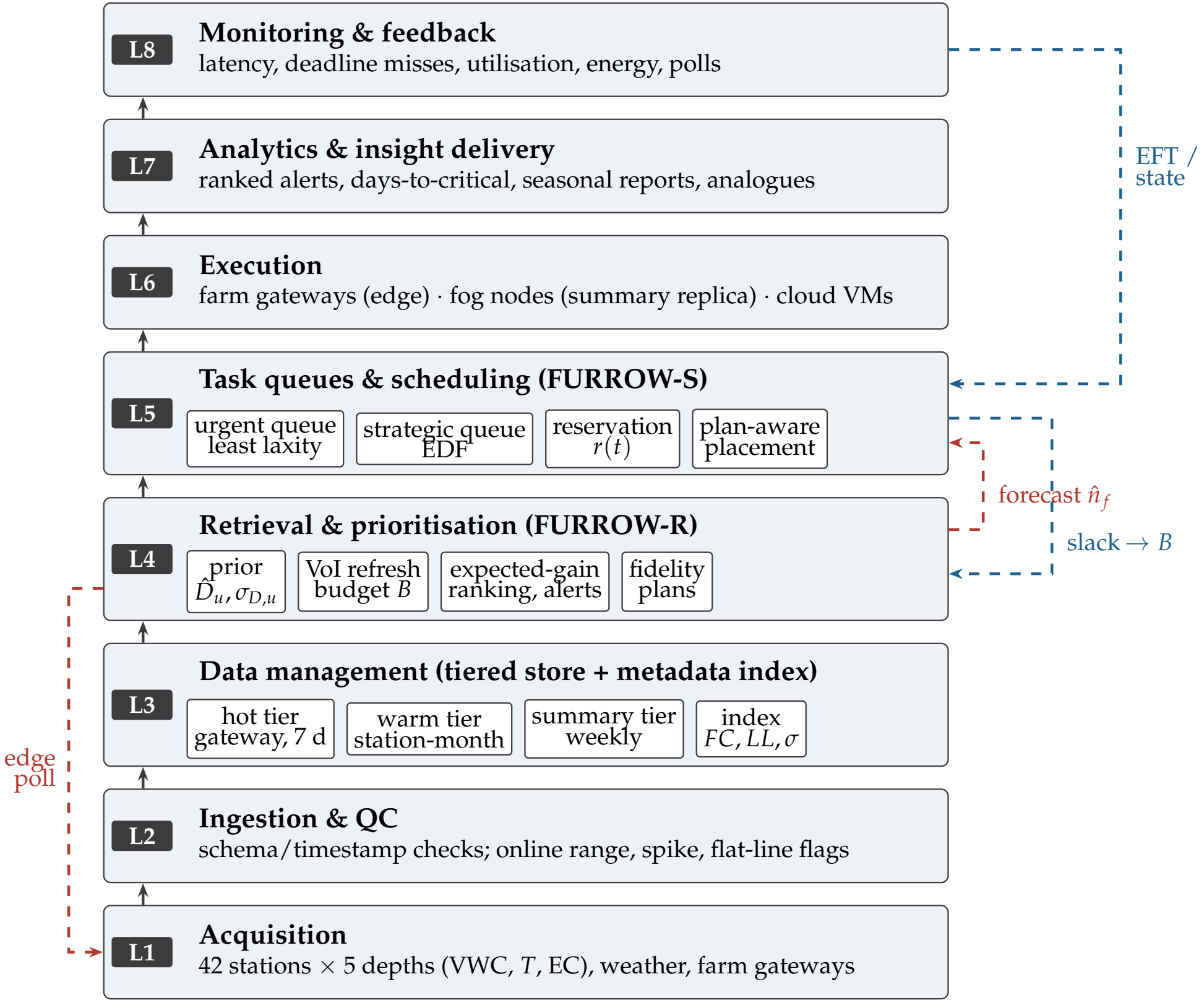

L8
Monitoring & feedback
latency, deadline misses, utilisation, energy, polls
L7
Analytics & insight delivery
ranked alerts, days-to-critical, seasonal reports, analogues
L6
Execution
farm gateways (edge) · fog nodes (summary replica) · cloud VMs
L5
Task queues & scheduling (FURROW-S)
urgent queue least laxity
strategic queue EDF
reservation $r(t)$
plan-aware placement
L4
Retrieval & prioritisation (FURROW-R)
prior $\hat{D}_u, \sigma_{D,u}$
VoI refresh budget B
expected-gain ranking, alerts
fidelity plans
L3
Data management (tiered store + metadata index)
hot tier gateway, 7 d
warm tier station-month
summary tier weekly
index FC, LL, σ
L2
Ingestion & QC
schema/timestamp checks; online range, spike, flat-line flags
L1
Acquisition
42 stations × 5 depths (VWC, T, EC), weather, farm gateways
EFT / state
forecast $\hat{n}_f$
slack → B
edge poll